

CAN'T SEE THE FOREST FOR THE TREES



COMPLEXITY, POWER, AND THE DUTY TO SEE
THE WHOLE IN THE AGE OF AI

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BUSINESS & STARTUPS

CAN'T SEE THE FOREST FOR THE TREES

*Complexity, Power, and the Duty to See the Whole in
the Age of AI*

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*The modern world rewards us for tending the tree. Almost
nobody is paid to ask what the forest is becoming.*

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaic. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

A NOTE ON METHOD

This is a human-directed, AI-assisted book. I chose the questions, arguments, examples, revisions, and final claims. AI was used for literature search, comparison, drafting, counterargument, and editing. That matters because one of the book's own warnings is that AI can produce coherence faster than truth can be checked. A smooth synthesis is not the same thing as a validated theory.

The work grew around Axiologic Research, Outfinity Venture Studio, and research activity connected to ACHILLES, but it is not an official project deliverable and it is not a catalogue of technologies. The books and technical experiments belong to the same larger inquiry: how humans can remain capable of orientation, correction, and sovereignty as AI makes local intellectual work cheaper and systems more complex.

Outfinitism is used here as a provisional cultural and research direction - a possible philosophy and science of limits, not a finished doctrine. The empirical claims that carry weight are referenced. The wider synthesis remains an argument to test. I expect some connections to fail. I would rather leave those limits visible than manufacture certainty.

Many books in the programme are published freely. That is deliberate. They are public intellectual experiments: invitations to criticize, reuse, falsify, and build. If they are ignored, they remain an archive. If they are useful, the useful parts should survive without requiring loyalty to the vocabulary that produced them.

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CHAPTER 1

The Argument in Ten Minutes

For roughly twenty-five years - essentially my entire adult life - I have kept returning to the same family of questions. Why do intelligent people build systems that later trap them? Why do institutions created to solve real problems become unable to see the problems created by their own solutions? Why does progress so often produce both freedom and dependence, clarity and blindness, resilience and fragility? I expected that one day I would write a book about this. I did not expect that trying to write it would become part of the evidence.

The immediate trigger was much smaller. While revising an earlier version of this manuscript, I asked an AI system to keep the urgency, doubt, and personal vulnerability of what I was saying. I wanted the text to admit something uncomfortable: after spending years around technology, research, startups, institutions, and AI, I am less certain than I used to be that our societies understand the systems they are building. I also wanted to admit the opposite: I still choose optimism, because the alternative is to stop looking for useful interventions before we have proved that none exist.

The AI improved the prose. It also weakened the point.

Nothing it wrote was obviously false. That was precisely the problem. It made the argument balanced, polite, and easy to accept. It removed some of the sentences that made me sound uncertain, some that made the situation sound urgent, and some that placed responsibility on people who actually make decisions. It converted a conflict into a reasonable discussion among compatible considerations. Sentence by sentence, the result was defensible. As a representation of what I meant, it was less true.

I call this smoothing. The term is provisional. I use it for a familiar social operation: remove enough sharp edges from a statement that incompatible audiences can accept it without confronting the disagreement underneath. Smoothing can be civilizing. It can reduce needless hostility, translate between vocabularies, and keep cooperation possible when complete agreement is impossible. But it can also become a subtle form of lying by omission. The hard part disappears while the words remain correct.

AI did not invent this. It learned it from us.

For years I described parts of modern public culture as hypocritical. I now think that word is often too simple. Hypocrisy suggests that a person knows one thing and says another. Much of what I see is more sophisticated. People inhabit several roles, incentives, loyalties, and vocabularies at once. A manager may sincerely care about employees and sincerely optimize a metric that harms them. A politician may sincerely defend democratic norms and sincerely use every legal advantage that weakens those norms when the other side seems threatening. A researcher may sincerely value truth and sincerely avoid a question that threatens publication, funding, or status. The contradiction is not always hidden inside one cynical mind. It can be distributed across a system.

This is why the American saying in the title matters: can't see the forest for the trees. Modern civilization is extraordinarily good at tending trees. The founder tends a company. The investor tends a portfolio. The scientist tends a problem. The engineer tends a subsystem. The civil servant tends a mandate. The lawyer tends a category of rights or liabilities. The journalist tends a story. The citizen tends a family, a career, and a flood of immediate demands. Many of these trees are beautiful. Some are profitable. Some are morally important. Some keep people alive.

The problem is not the trees. The forest is made of trees.

The problem is that very few roles are rewarded for asking what all the trees are doing together.

That failure is not merely philosophical. It is becoming a business, political, and technological risk. Complexity has grown into one of the main resources of the modern world. We create value by mastering domains that other people cannot easily master: semiconductor fabrication, derivatives, molecular biology, cybersecurity, regulation, machine learning, logistics, tax law, standards, platform governance. Specialization is productive because knowledge is too large for one mind. But the same complexity creates rents, prestige, barriers to entry, dependence, and opportunities for obfuscation. There is a real economic literature on strategic complexity: firms can make offers harder to compare; managers can make disclosures harder to read; competition itself can create incentives to complicate choice (Spiegler, 2016; Edwards, 2019; Aghamolla and Smith, 2023).

Complexity therefore does two things at once. It allows civilization to become more capable, and it makes the consequences of capability harder to integrate.

This tension resembles, very loosely, the old yin-yang intuition that one tendency can create conditions for its apparent opposite. I do not treat yin-yang as a scientific law. Attempts to map the philosophy directly onto measurable biological pairs have not established a universal empirical theory, and recent reviews still note major gaps in mechanism and quantification (Wang, 2005; Yang et al., 2025). I use it as a warning against one-directional thinking. More local sophistication can coexist with less global comprehension. More information can coexist with less orientation. More connectivity can coexist with deeper social isolation. More control can create more fragility. More speech can make important speech harder to hear.

The last example matters because modern censorship can operate without a censor. I am not using censorship here in the legal sense. Nobody needs to ban a book for the book to become functionally absent. Put it inside an almost infinite competition for attention and it can disappear more effectively than if it were prohibited and therefore made interesting.

The incentive structure is not neutral. Large studies of social media have found that political out-group language predicts sharing unusually strongly, that social feedback reinforces future expressions of moral outrage, and that false stories in one major Twitter dataset spread farther, faster, deeper, and more broadly than true ones (Rathje, Van Bavel, and van der Linden, 2021; Brady et al., 2021; Vosoughi, Roy, and Aral, 2018). A preregistered audit published in 2025 found that Twitter's engagement-based ranking amplified emotionally charged and out-group-hostile political content relative to a reverse-chronological feed, including content users themselves did not say they preferred (Milli et al., 2025). Evidence on echo chambers is more mixed than public rhetoric suggests, but the larger point survives the nuance: visibility is allocated by mechanisms that reward measurable response, and measurable response is not the same thing as considered value.

This creates a strange information regime. A crude statement can travel because it irritates. A false statement can travel because it is novel. A tribal statement can travel because it activates identity. A serious argument that asks the reader to hold three conflicting mechanisms in mind may lose before anyone evaluates whether it is true.

That is the environment in which this book and the other free books around it exist.

I did not begin with a plan to publish a shelf of books. I began with philosophy and literature, and with a persistent interest in limits. Over the years that interest became what I now call Outfinitism: not a finished doctrine, but a way of looking at the world that starts from a simple fact. Every useful model has a domain. Every domain has an edge. The fact that a model works brilliantly inside its domain gives it no automatic authority outside it.

The deeper substrate is meta-rationality. The map is not the territory, but that sentence is useless if it becomes an excuse for cheap relativism. Some maps are better than others for a given purpose. Some theories predict well in one regime and fail in another. Two theories can contradict each other and still be useful at different scales or under different conditions. The practical task is not to dissolve truth into perspectives. It is to develop a better discipline for knowing where a theory, metric, institution, or AI system stops deserving our confidence.

When our research programme began using AI to examine a few aspects of this problem, I thought the work would converge. We would write several books, clarify the main ideas, connect them to technical experiments and perhaps to literature, and gradually reduce the backlog of unanswered questions.

The opposite happened.

Each answer exposed another boundary. AI agents raised questions about authority. Authority raised questions about permission, memory, accountability, and sovereignty. Scientific automation raised questions about what counts as evidence, review, novelty, and explanation. Institutions led to power. Power led to attention. Attention led to trivialization, preference falsification, and the economics of outrage. Social systems led to suffering, adaptation, mental health, status, and the strange ways people remain functional inside structures they privately distrust. Literature became useful because some conflicts are easier to perceive when they are inhabited by characters than when they are compressed into a diagram.

The collection grew because the world did not respect our chapter boundaries.

I am not claiming that this makes me unusually wise. A less flattering explanation is probably more accurate. I have been lucky enough to spend a large part of my adult life obsessed with coherence and with things that do not fit. I notice cognitive dissonance and incompatible explanations and have difficulty leaving them alone. AI then gave that obsession leverage. It became possible to

follow one inconsistency into another discipline without waiting months for the logistics of research and drafting.

Obsession is not proof.

AI also creates a new epistemic danger. A language model is very good at connecting points. Sometimes the connection is real. Sometimes coherence pressure produces the impression of an argument where the evidence only supports adjacent observations. A thousand elegant links can still inherit one hidden mistake. Evidence is correlated; quantity is not independence. The books therefore cannot be treated as a proof that Outfinitism, or any larger diagnosis, is correct. They are a research programme, a set of experiments, and a growing collection of claims that should survive criticism separately before they are allowed to support a grand conclusion.

Still, the pattern is now too persistent for me to dismiss as one hallucinated connection. We keep finding the same structural problems in different places: local optimization without integration, metrics that become targets, interfaces that hide dependency, institutions that accumulate rules faster than causal memory, markets that price what can be captured and neglect what cannot, political systems that compress millions of lives into administrable categories, and information platforms that can measure reaction more easily than reflection.

The first claim of this book is therefore modest in wording and severe in consequence: humanity's capacity to create complexity is growing faster than the unaided human capacity to understand and govern the systems created.

The second claim is biological. Human attention, working memory, time, and emotional energy are limited. Training can improve what we do with them; it cannot make them infinite. Expertise works partly by compression. The expert sees patterns that the novice must inspect one element at a time. But every compression discards something. Herbert Simon's bounded rationality remains a better starting point for real institutions than the fantasy of an actor who sees all options and consequences (Simon, 1955).

The third claim is about power. Power is partly the right to simplify reality for other people. A state defines categories. A company defines KPIs. A market turns futures into prices. A platform ranks what is visible. An investor defines what counts as a fundable company. An AI system selects context before generating an answer. The simplification is often necessary. Whoever controls it also controls part of what can be noticed.

The fourth claim is about AI. If AI increasingly replaces local intellectual labour - drafting, coding, search, comparison, simulation, summarization, routine design - then the human scarcity moves upward. The difficult work becomes choosing the frame, seeing its edge, orchestrating processes across domains, deciding which contradiction cannot be smoothed away, and accepting responsibility for consequences that no local optimization owned.

This makes the work that motivated Outfinitism more relevant, not less. The goal is not to produce humans who know every leaf. It is to create habits, tools, institutions, and forms of education that make scale-switching normal: look closely when detail matters, look up when the local objective begins to damage the surrounding system, and state the boundary when the model no longer deserves authority.

I do not know whether this message will break through the attention barrier. That uncertainty belongs in the book. We have already discovered that making serious books free does not make them visible. Promotion is difficult precisely because the argument resists compression into one tribal slogan. It may be ignored. It may be read by a small group. It may become useful only after some of the systems discussed here fail in ways that make the vocabulary suddenly relevant.

That possibility is not an excuse for passivity. Prevention has an unfair business model: when it succeeds, the disaster that would have justified the investment never occurs. The people who remove resilience can look efficient until the shock arrives. The people who preserve it can look wasteful during every quarter in which nothing breaks.

I increasingly think this is one of the central moral problems of leadership in complex systems. We are rewarded for tending the visible tree. We are rarely rewarded for preserving the conditions under which the forest survives.

The rest of the book develops this argument from several directions: cognition, organizations, startups, markets, media, political systems, AI, investment, and the philosophy of limits. I will try not to smooth the conflicts into one happy synthesis. Some tensions are real. Democracy is both slow and resilient. Centralization is both efficient and dangerous. Complexity is both the source of modern capability and a medium through which power hides. AI is both an amplifier of confusion and the best tool we have yet built for extending the range of human inquiry.

A compact map of the argument

Observation	What it demands
Local intelligence becomes abundant	Move attention from production alone to selection, integration, and responsibility.
Human cognition remains finite	Design for attention limits; no dashboard abolishes biology.
Complexity creates value and rents	Distinguish useful specialization from dependency, obfuscation, and prestige games.
Power chooses simplifications	Make consequential maps visible, contestable, and revisable.
Attention is a selection system	Do not confuse engagement with considered value.
AI accelerates feedback	Preserve provenance, dissent, reversibility, and the ability to stop.
Outfinitism focuses on edges	State domains, frontiers, invalidators, and recovery paths.
Prevention is under-rewarded	Fund resilience before pain creates an obvious market.

My own position will continue to oscillate. The pessimistic case is strong. Intelligent people repeatedly fail in front of complexity, and AI is shortening the time between invention and consequence. The optimistic case is weaker as a prediction but stronger as a choice: systems can learn, people can build better institutions, distributed intelligence can outperform brittle central control, and exposing the limits of our maps can make correction possible earlier.

I did not start the experiment that is human history. None of us did. But I am inside it, benefiting from it, building technologies within it, and therefore partly responsible for what I can see.

That is the call to action in its simplest form. Tend your tree. Do it well. Then look up.

CHAPTER 2

Twenty-Five Years at the Edges

I can trace the beginning of this book back much further than the current AI wave. For about twenty-five years I have moved through software engineering, cybersecurity, privacy, blockchain, decentralized systems, artificial intelligence, European research, startups, and literature. The technologies changed. The irritation did not.

Again and again I encountered systems that were intelligent inside their own borders and surprisingly stupid at the border itself. A security architecture protected the component and ignored the organization that would operate it. A privacy mechanism was mathematically elegant and unusable in the institutional process around it. A regulation solved one historical abuse and created a new dependency that nobody owned. A business case optimized a market transaction and treated the social infrastructure making that transaction possible as scenery.

The pattern was not that experts were incompetent. Usually the opposite was true. They were competent enough to build deep local models, and the organization rewarded them for staying inside those models. The fracture appeared where several valid descriptions met and no role had authority to integrate them.

For years I thought the right response was a book. One sufficiently broad book could perhaps connect technology, institutions, power, and the limits of human reasoning. The ambition was naive, but useful. It forced me to ask what such a book would need to explain.

Outfinitism emerged slowly from that question. The word is intentionally awkward. It points outward, toward the point where a framework stops. I do not claim a complete philosophy under the name. I use it for a cultural direction: make limits visible, test them, and design systems that can remain corrigible when successful models become institutions.

The meta-rationalist part matters most. Rationality inside one frame is not enough when reality contains several partially incompatible frames. A market model, a legal model, an ecological model, and a psychological model can each be internally coherent and still recommend incompatible actions. The answer is not to pretend that all are equally true. It is to learn the conditions under which each

is informative, what it leaves out, and what happens when its recommendations cross scale.

When generative AI became useful enough to support serious research and long-form drafting, a practical opportunity appeared. Instead of waiting years to write one synthesis, we could examine the problem from several directions. Some work grew around research projects, especially the environment created by ACHILLES. Some grew from Axiologic Research. Some belonged to Outfinity and the attempt to connect intellectual work to sustainable ventures. Some was simply curiosity that refused to remain inside a work package.

The first books were meant to explore a few aspects of Outfinitism and related questions. Then the project escaped its plan.

Nothing mystical happened. I still selected questions, rejected drafts, changed titles, argued with the AI, and decided what was worth publishing. What changed was the shape of the problem. Each useful distinction exposed another boundary beside it. One book made the need for another harder to ignore.

A question about trustworthy AI became a question about who remains responsible when an agent acts. That became a question about sovereignty over memory, evidence, permissions, and infrastructure. A question about scientific automation became a question about whether research can become more inspectable and partly machine-interpretable without becoming ritualized. A question about social systems became a question about suffering, adaptation, mental illness, elites, envy, status, attention, and the difference between a society that looks stable and one that is merely good at hiding its contradictions. Several books ended up asking the same uncomfortable question from different sides: what kind of person does a system train us to become in order to remain functional inside it?

The technical work and the books were never separate in spirit. The technology asks how to make AI-mediated work more traceable, portable, inspectable, and correctable. The books ask why those properties matter, where they might fail, who gains power when they disappear, and what kind of human life remains possible when machines become better at local intellectual work.

The technical intuition is general. In a world of increasingly capable models, the durable object should be the human project: its evidence, history, permissions, values, decisions, and ability to move between providers and models. AI should

extend the project without quietly becoming the owner of its memory or the hidden authority behind its decisions.

The same idea explains why many of the books are free. Free access is not charity theater. It is part of the experiment. If the work is trying to expose assumptions and build a vocabulary for problems that do not yet have a market, locking the vocabulary behind a price barrier weakens criticism and slows diffusion. The economic return, if one appears, may come through research, ventures, collaboration, standards, technologies, or ideas that become useful years later.

There is also a less strategic reason. Some questions deserve serious treatment before demand exists. Markets are excellent at discovering many forms of value. They are structurally weak at rewarding benefits that remain counterfactual, diffuse, public, or too early to fit a purchasing category. A book that helps someone avoid a mistake five years later cannot easily invoice the avoided mistake.

Free, however, is not visible.

This has been one of the more sobering lessons of the programme. I can put a serious book online for anyone in the world to download, translate it, link it, and explain why I think it matters. It can still disappear. The attention barrier is often harder than the price barrier.

At first I took that personally. Then I realized it was another example of the thesis. The modern information environment does not rank claims by civilizational importance. It cannot. Platforms optimize measurable signals. Readers optimize finite time. Journalists need stories. Investors need categories. Researchers need literatures. Everyone is solving a local selection problem under scarcity.

The catalogue therefore has two possible futures, and I want both stated honestly. In one future it becomes a useful intellectual infrastructure: founders, investors, researchers, policymakers, and writers reuse its distinctions, falsify weak ideas, and turn stronger ones into institutions or technologies. In another future it remains mostly unread until events make some of the questions obvious.

The second future would be disappointing, but not worthless. Archives matter after systems fail. A society rebuilding after a crisis does not start from zero; it starts from whatever concepts, skills, institutions, records, and prototypes survived. If some of these books become witnesses - evidence that certain fractures

were visible before they became emergencies - they can still reduce the intellectual cost of reconstruction.

I would prefer prevention. I am not romanticizing collapse. The fact that reconstruction can use old ideas is no reason to welcome the destruction that makes them relevant. But a programme about limits should admit its own. We may not be able to create demand for prevention before the danger becomes concrete. We may not be able to make nuanced arguments competitive in attention markets built around faster rewards. We may discover that some of our own theories are wrong.

The programme as a set of experiments, not a doctrine

Line of inquiry	Question it keeps reopening
Limits and meta-rationality	Where does a useful model stop deserving confidence?
Trustworthy and delegated AI	Who remains responsible when machines perform local intellectual work?
Digital sovereignty and continuity	Can a human project remain portable, inspectable, and governable across models and providers?
Inspectable research	Can evidence, uncertainty, and reasoning remain traceable as research becomes more automated?
Institutions, power, and attention	Who chooses the map, who benefits from complexity, and what becomes invisible?
Suffering and adaptation	What kind of person does a system train us to become in order to function inside it?
Business and venture experiments	Which ideas survive contact with implementation, incentives, customers, and capital?

That is why I increasingly see the collection not as a doctrine but as a scaffolding. Some beams will be removed. Some connections will be wrong. Some books will age badly. The useful part is the habit: treat the world as a system of systems; ask where the frame ends; preserve enough disagreement that later readers can see what was uncertain rather than inheriting a polished retrospective certainty.

The project began because I wanted to write about the limits of our maps. It expanded because every map had an edge. The irony is now obvious: even the map of the edges became too large for one person to hold comfortably in mind.

That is not a victory. It is the problem.

CHAPTER 3

A World of Carefully Tended Trees

Modern civilization is built by specialists. This is one of its greatest achievements and one of its deepest vulnerabilities.

A product team improves conversion. A laboratory narrows a mechanism. A ministry administers a mandate. A fund refines an investment thesis. A hospital optimizes a clinical pathway. A platform improves ranking. A security team closes vulnerabilities. Each action may be rational. The danger appears when the local success measure is allowed to stand in for the health of the system that contains it.

Organizational research has a useful name for part of this: coordination neglect. People are often better at dividing a task than at reintegrating the pieces, and they tend to focus on components even when the components are tightly interdependent (Heath and Staudenmayer, 2000). Specialization solves information problems by reducing what each person must know. It then creates an integration problem that no specialist can solve from inside the specialty.

This is the forest problem in institutional form.

I do not believe the remedy is a cult of generalists. A person who knows a little about everything can be dangerous when confidence grows faster than depth. The system needs deep trees and people who know them intimately. It also needs interfaces, translators, adversarial reviewers, integrators, and leaders who can change scale before a local optimum becomes a global trap.

The difficulty is incentive alignment. The product manager is reviewed on the product. The fund manager is reviewed on returns. The scientist is reviewed on research output. The minister is reviewed within a political horizon. The person who says, "our local goal is working but the surrounding system is deteriorating" often has no metric, no budget, and no clear authority.

That absence is not accidental. The whole is harder to measure than the part. The part has an owner. The externality does not.

Complexity intensifies the problem because the modern world is not merely large. It is adaptive, coupled, delayed, multi-scale, and reflexive. People react to rules. Markets react to predictions. Adversaries react to controls. Models change the behavior that generates future data. A useful intervention at one scale can become destructive at another.

Agriculture did not only produce more food. It produced storage, property, taxation, hierarchy, records, warfare over surplus, and institutions to stabilize all of these. Industrialization did not only produce machines. It reorganized cities, labor, education, finance, family life, health, and the meaning of time. Digital networks did not only produce communication. They became infrastructure for identity, payment, reputation, entertainment, politics, surveillance, work, and memory.

Each interface became simpler for the user by moving complexity somewhere else.

That sentence deserves more attention. Technology often gives us the experience of simplification while increasing the complexity of the surrounding dependency graph. A tap on a screen can trigger cryptography, identity systems, cloud services, payment rails, telecom networks, logistics, legal contracts, energy use, standards, and geopolitical supply chains. The user sees one button. The system acquires another layer of dependence.

This is not an argument against the button. It is an argument against confusing interface simplicity with system simplicity.

Herbert Simon's work on nearly decomposable systems explains why modularity is so powerful. Strong interaction inside modules and weaker interaction across modules lets each part be understood without reconstructing the entire system (Simon, 1962). Software, firms, markets, and institutions all use forms of modularity to make complexity manageable.

But modularity moves danger to the seams. The assumptions between components become critical. A supplier can be reliable and the shared dependency fragile. Two departments can meet their targets while the handoff fails. A model can be correct about the variable it predicts and useless for the decision made from the prediction.

I find four recurring ways that complexity defeats intuition: scale, coupling, adaptation, and delay. At scale, relationships change. Under coupling, disturbances travel. Under adaptation, people reorganize around the rule. Under

delay, the person receiving the benefit and the person paying the cost may live in different quarters, governments, or generations.

Four recurring ways complexity defeats intuition

Form	Failure of local reasoning
Scale	A relationship that works locally changes when the system becomes larger or more concentrated.
Coupling	A disturbance travels through dependencies that no local owner fully sees.
Adaptation	People, firms, adversaries, and models change behavior in response to the rule.
Delay	Benefits arrive now while costs surface elsewhere, later, or for someone else.

These mechanisms make complexity economically valuable. Whoever can absorb complexity for others can capture value. Platforms coordinate millions of users. Standards make incompatible systems interoperable. Consultants translate domains. Financial products compress future uncertainty. AI systems summarize more information than a person could read.

But complexity can also be manufactured and defended. Economic models of obfuscation show that firms may deliberately make comparison harder when consumers have limited attention or search capacity. Financial-disclosure research finds circumstances in which complex language is used strategically around bad news, though complexity can also carry legitimate information (Spiegler, 2016; Aghamolla and Smith, 2023). The point is not that every complex document is dishonest. The point is that complexity itself can become part of competition.

Social prestige can work similarly even when nobody plans it. Mastery of a difficult vocabulary signals membership. The vocabulary may be necessary; every mature field needs compression. But once status attaches to the code, clarity can threaten the status structure. An institution can start rewarding the ability to navigate its complexity more than the ability to ask whether the complexity still serves its purpose.

That is where the word bureaucracy becomes too easy. Many rules encode memory. A building code can be the fossil of a fire. A clinical protocol can preserve a lesson no current employee personally witnessed. A privacy rule can

exist because an earlier generation discovered what powerful institutions do when data flows without constraint. Removing a rule can remove memory while leaving the hazard.

The hard work is not simplification. It is intelligent simplification: preserve the causal memory, remove ritual, make exceptions visible, and keep the system capable of learning when the environment changes.

This is also why I resist simple anti-elite narratives. Complex societies need people and institutions able to integrate knowledge across domains. Intellectual, economic, administrative, and technical elites have always influenced the direction of societies because they control resources, vocabularies, institutions, and coordination capacity. The problem is not the existence of expertise or leadership. The problem is when the map-makers cease to be correctable by the territory.

A society of carefully tended trees can remain healthy if someone maintains the soil, water, firebreaks, biodiversity, and paths between them. It becomes dangerous when every budget line belongs to a tree and the forest itself is treated as an abstraction.

CHAPTER 4

A Brain Built for Another Scale

The forest problem is not only institutional. It is biological.

We are asking human beings to navigate systems with millions of actors, global supply chains, continuous information flows, overlapping legal regimes, algorithmic intermediaries, rapidly changing technologies, and consequences delayed across years. The brain doing the navigation is the product of a very different evolutionary history.

This observation can be abused. It is easy to turn "humans are limited" into a lazy story about caveman brains or into an excuse for paternalism. Human cognition is extraordinarily flexible. Education, writing, mathematics, institutions, science, markets, and computers are all technologies for extending it. The point is not that we are doomed by biology. The point is that every extension still operates through scarce attention, time, energy, and motivation.

Research on working memory and attention does not support the idea of an unlimited general workspace. Estimates vary by task and representation, but the central fact is robust: only a limited amount can be actively maintained and controlled at once (Cowan, 2001). Reviews of attention emphasize competition for limited representational resources; mental effort itself is allocated in relation to cost and reward (Scalf et al., 2013; Shenhav et al., 2017).

Expertise works by changing what counts as one thing. A chess master does not see thirty-two unrelated pieces. A programmer sees a pattern. A physician sees a syndrome. A lawyer sees a doctrinal structure. Compression lets limited cognition cover more territory.

Compression is also where the loss begins.

The expert sees faster because some distinctions have become irrelevant to the expert's task. Change the task and the discarded distinction may suddenly matter. This is why deep expertise can coexist with surprising blindness outside its trained frame. It is not stupidity. It is the cost structure of competence.

Division of labor scales the same logic across society. We place fragments of knowledge into different people and institutions and coordinate them through prices, hierarchies, standards, professions, and protocols. The intelligence of civilization becomes distributed. No one understands a smartphone, a hospital, a

financial system, or a modern state in full. Yet the system works because each person can rely on abstractions maintained by others.

Until the abstractions fail.

One reason I keep returning to the forest metaphor is that it makes the cognitive trade-off visible. The better I become at reading the leaves of one tree, the more time I spend away from the ridge from which the forest can be seen. Training can improve both local and global understanding, but the hours are still the same hours. Family, health, income, friendship, administration, entertainment, and simple fatigue compete for them too.

The trade-off is not mathematically fixed, but it is real enough to organize institutions around. A civilization that wants people to see larger patterns must create time, education, tools, and status for integration. It cannot simply tell exhausted specialists to "think systemically" after filling every hour with local deliverables.

This is where the yin-yang analogy becomes useful to me, provided it stays an analogy. As local capability grows, the potential for global blindness can grow with it. The more we can do through specialization, the more dependent we become on integration. The more powerful the instrument, the more expensive the error in choosing the wrong frame.

Taoist yin-yang language is not a substitute for empirical explanation. Historical attempts to identify a universal measurable essence behind yin and yang have not established such a theory, and even sympathetic recent reviews acknowledge unresolved mechanistic and quantitative problems (Wang, 2005; Yang et al., 2025). What the metaphor contributes is an intellectual reflex: look for the counter-tendency generated by success.

Efficiency can create fragility by removing slack. Transparency can create performative behavior when every signal becomes a target. Connectivity can create common-mode failure. Personalization can create dependency on an intermediary. Safety controls can create complacency. The relationship is not mystical; each case has a mechanism.

The modern attention economy adds another layer. Our cognitive limits are not merely constraints to be respected. They are resources over which firms compete. The notification, headline, recommendation, outrage cycle, infinite scroll, and personalized feed are all technologies for winning a scarce allocation decision inside the mind.

This does not make users helpless. People choose, learn, resist, and create norms around media. But the competition is asymmetric. A person tries to preserve attention while thousands of optimized systems test what makes attention move.

The predictable result is not universal stupidity. It is fragmentation. A person can be brilliant at work, careful with family, politically informed, physically exhausted, and unable to devote three uninterrupted hours to a difficult argument about a system whose consequences will arrive in ten years.

This is why I reject the flattering version of my own story. I do not see more because I stand above other people. I may simply have spent an unusual amount of time on a narrow obsession: contradictions between systems. That obsession has costs and blind spots of its own. AI has made it possible to extend the range, but it has not removed the need to decide which connections are real.

The practical question is therefore not how to create omniscient citizens. It is how to build a civilization in which finite citizens can remain sovereign enough to challenge the systems they depend on.

That requires cognitive prosthetics that reveal rather than hide limits, institutions that preserve dissent, education that teaches scale-switching, and interfaces that allow progressive disclosure instead of choosing between one green light and a thousand-page manual.

The brain was built for another scale. Civilization is the machinery by which we extend it. The next question is who controls that machinery.

CHAPTER 5

Power Is the Right to Simplify

Every system of power is also a system of representation.

A state cannot administer millions of unique biographies. It creates categories. A corporation cannot manage every interaction directly. It creates metrics, roles, budgets, and dashboards. A market cannot carry every motive inside a transaction. It creates prices. A platform cannot show everything. It ranks. An investor cannot model every possible future. A thesis filters the space.

These compressions are not optional. Without them, coordination collapses under its own informational weight.

Power enters because someone chooses the compression.

A metric is a promise that one observable number stands for something we care about. Revenue stands for exchange. Growth stands for expanding relevance. Publication count stands for research activity. Response time stands for service quality. Test scores stand for learning. None is absurd. None is the thing itself.

Campbell's law and the family of ideas usually associated with Goodhart's law describe what happens when the representation becomes a target. People adapt to it. A school teaches to the test. A sales organization books contracts that another department will later regret. A hospital avoids cases that damage a performance indicator. A research system rewards publication and receives behavior optimized for publishability rather than for epistemic risk (Campbell, 1976).

The problem is not measurement. Refusing to measure can protect incompetence just as easily as excessive measurement can create gaming. The problem is forgetting what the measure excludes after resources and status have been attached to it.

Power compresses reality

Compression	What the map can hide
State categories	People who do not fit the administrative class.
Corporate KPIs	Costs shifted outside the metric or reporting period.
Prices	Public goods, power, dignity, and unpriced externalities.

Compression	What the map can hide
Votes and polls	Intensity, ambiguity, future preference, and non-participation.
Platform rankings	Value without immediate measurable response.
AI context selection	Evidence and contradictions omitted before synthesis.

The shadow around a compression is where power becomes difficult to contest. The employee knows the KPI is wrong but is evaluated by it. The citizen knows the administrative category does not fit but must enter it to access a service. The creator adapts to a ranking system whose objective is proprietary. The startup tells its story in the categories through which capital recognizes opportunity.

Complexity can protect these arrangements. A system can become too difficult for outsiders to challenge and too distributed for insiders to own. Sometimes this is unavoidable. Financial systems, AI models, tax codes, and medical infrastructures genuinely are complex. Sometimes complexity becomes strategic. The literature on obfuscation is useful precisely because it shows that opacity need not be an accidental cost of sophistication; it can be profitable (Spiegler, 2016; Edwards, 2019).

This gives complexity a second economic role. It is not only a cost to be reduced. It can be a moat.

The person or institution able to navigate the maze has an advantage. Lawyers, consultants, regulators, engineers, financiers, and platform operators become translators between the formal system and everyone who must live inside it. There is nothing inherently wrong with that. Expertise deserves compensation. The danger appears when the maze has no independent reason to remain a maze except that too many positions now depend on navigation being difficult.

The same applies to intellectual prestige. A field needs technical language because ordinary language cannot carry every distinction efficiently. But jargon can cross a frontier where it stops compressing knowledge and starts screening membership. I have seen this in technology and research, and I have certainly contributed to it. Complexity can make us feel serious even when the underlying proposition is simple.

A book about complexity should not use complexity as camouflage. If a claim can be stated in three sentences, ten pages do not make it more true.

Hayek's argument about dispersed knowledge remains one of the clearest demonstrations of why centralized maps fail. Economically relevant knowledge is local, practical, changing, and often impossible to transmit in full. Prices coordinate fragments without requiring a planner to possess them all (Hayek, 1945). The insight is real.

The mistake is to turn one powerful compression into a universal language. Prices do not automatically represent harms imposed outside the transaction, inability to pay, rights, meaning, or irreversible losses. They are excellent instruments inside a frame. They become ideology when the frame disappears.

Voting is another compression. Millions of reasons become offices and mandates. Democracy is not made legitimate because the vote preserves every preference. It is made more legitimate because the vote sits inside a corrective ecology: rights, opposition, courts, media, local institutions, associations, experts, procedures, and the possibility of another vote.

That ecology matters more in the age of AI because AI can become a general-purpose compression layer. It can summarize the report before the executive reads it, rank applicants before the manager meets them, suggest the diagnosis before the physician decides, generate the options before the cabinet debates, and select the evidence before the researcher writes.

Formal authority may stay human while practical visibility becomes machine-mediated.

The political question is therefore often less dramatic than "will AI seize power?" Ask instead: who chooses the context? Who defines the objective? Which data persists? Which uncertainty is visible? Which alternatives are never generated? Who can appeal when the machine's compression becomes infrastructure?

This is where my objection to euphemistic power becomes simple. Ambition is not the problem. If you want to be king of the world, say so. Then other people can evaluate the crown, the cost, and the legitimacy of your attempt. The destabilizing strategy is to accumulate control while speaking a language in which nobody is controlling anything, every concentration is merely convenience, and every asymmetry is described as an unfortunate implementation detail.

That style can be locally adaptive. It reduces opposition. It preserves coalition. It is also weak. A system that cannot name its own power cannot examine the failure modes of that power. The lie returns as fragility.

A mature system does not need to reveal every internal detail to everyone. It needs contestability proportional to consequence. The more a compression controls access, safety, reputation, money, liberty, or political visibility, the more important it becomes that someone can inspect its boundary and force revision.

The map will never become the territory. The practical question is whether the territory still has a way to complain.

CHAPTER 6

The Entrepreneur's Necessary Blindness

Entrepreneurs are paid to ignore most of the world.

That sentence sounds like criticism. It is mostly praise.

A startup cannot begin by solving civilization. It needs a customer, a problem, a product, a price, and a feedback loop. The discipline of entrepreneurship is to cut away abstraction until somebody cares enough to use or buy what has been built. Compared with committees that spend years defining a problem and never touch reality, this narrowness is a form of honesty.

The trouble begins when a method for finding local truth is promoted into a theory of global progress.

A startup asks whether a user has pain. Society has to ask who is not a user. A startup asks whether retention improves. Society has to ask what dependence means after ten years. A founder asks whether the unit economics work. Society has to ask which public goods, environmental costs, legal protections, cognitive habits, and infrastructures sit outside the unit.

The startup is not morally required to solve every externality before writing its first line of code. If that were the standard, nothing new would be built. But scale changes the obligation. A product used by a thousand volunteers and the same product embedded in the daily behavior of a billion people are not the same intervention with a bigger database.

The founder's useful narrowness - and what it misses

Local optimization	Forest-level question
Acute user pain	Who benefits or pays without being a customer?
Customer feedback	Which effects arrive too late to appear in feedback?
Growth and retention	What happens to resilience and dependency at scale?

Local optimization	Forest-level question
Speed and experimentation	Which early choices become expensive to reverse?
Unit economics	Which public goods make the unit possible?
Competitive advantage	Does success create a common-mode failure or gatekeeper?

The most dangerous phrase in business may be "not our problem" when scale has made the problem inseparable from the product.

I have spent enough time around startups to understand why founders resist this language. Early companies die from lack of focus. Every new stakeholder can become an excuse for indecision. Consultants can turn a simple product into a governance theatre before there is anything worth governing. The correct lesson is not to replace speed with paralysis.

The lesson is to know when the frame has changed.

An early founder needs radical local focus. A scaled founder needs the ability to notice when the local optimization has become infrastructure for other people's lives. That transition should be treated as a change in kind, not merely a change in revenue.

Digital platforms offer the obvious examples because their externalities are visible. Engagement, recommendation, identity, communication, advertising, and moderation started as product problems and became elements of the public sphere. Similar transitions occur in finance, health, energy, logistics, cloud computing, and increasingly AI. Infrastructure inherits political characteristics whether its founders wanted politics or not.

The investor faces the same problem. Venture capital is structurally attracted to optionality early and concentration later. Many experiments are funded, a few winners scale, and returns come from the fact that the winner becomes unusually important. This is productive when the winning company absorbs complexity for society. It is dangerous when returns depend on transferring complexity or risk outside the transaction.

A useful investment question is therefore not only "what problem does this solve?" but "where does the unresolved part go?"

Does the product eliminate work or move it to unpaid users? Does the platform reduce transaction costs or create a private tax on an ecosystem? Does the AI system increase capability or centralize the memory and workflow of the customer? Does the low price come from genuine efficiency or from a liability that will mature after the company has been sold?

None of these questions has a universal anti-business answer. Sometimes concentration is the efficient architecture. Sometimes externalities are small. Sometimes a founder cannot solve the surrounding institutional problem and should not pretend otherwise. The point is to keep the ledger intellectually open.

This is where prevention collides with venture logic. Prevention is difficult to monetize because its output is an absence. A cyberattack does not happen. A supply chain does not fail. A democracy does not cross a threshold. A model does not create a scandal. A person does not develop a dependence that later becomes expensive.

The person who removes resilience can look efficient for years. The person who pays for redundancy looks wasteful until the day redundancy is needed.

Businesses know this problem in narrower forms. Insurance, security, maintenance, compliance, and quality assurance all sell some version of prevention. They become easier to fund when the probability and cost of failure are legible. Civilizational prevention is harder because the beneficiaries are diffuse, causal chains are contested, and the investor cannot easily capture the upside.

This creates a bias toward visible opportunity and away from invisible fragility. It also creates an opening for entrepreneurs. As complexity rises, products that improve orientation, verification, resilience, portability, privacy, provenance, and coordination can become markets in their own right. The fact that the market is early does not mean it is imaginary. It may mean the pain has not yet been priced.

I am interested in startups for exactly this reason. Business is not separate from philosophy. A venture is an executable theory about what people need, what they will pay for, what can be coordinated, and which dependencies can be tolerated. It exposes assumptions faster than an essay does.

But I do not want business to become the moral alibi of a system that knows only how to fund what can be captured quickly. If technology is accelerating the production of systemic risk, then capital has a responsibility to develop better categories for companies whose value includes prevention, sovereignty, correction, and resilience.

The founder's narrowness is necessary. The founder's permanence inside that narrowness is not.

A good entrepreneur should be able to say, "I am focusing on this tree because it is the only way to build anything," and later, "the tree is now large enough to alter the forest, so my responsibility has changed."

That scale-switch is one of the forms of maturity this book is asking for.

CHAPTER 7

The Attention Market

A book can be free, true, relevant, and almost invisible.

This is not a paradox. It is the normal condition of abundance.

The internet removed many distribution bottlenecks. Generative AI is removing production bottlenecks. Neither removes the biological bottleneck at the receiving end. Human attention remains finite, so abundance turns attention into a market even when no money changes hands.

I have felt this directly while trying to promote serious books. The problem is not merely marketing competence. A book about the interaction of AI, power, institutions, suffering, incentives, and long-term risk is competing against messages that can be understood in one second. The serious argument is often asked to pay a cognitive entry fee before the reader even knows whether the return will be worth it.

Platforms face the same selection problem at enormous scale. They need to decide what to show first. Engagement is attractive because it is measurable. A click, share, comment, watch time, or reaction is available immediately. Reflection, regret, learning, civic value, and whether the user is glad to have spent the last hour this way are harder to measure.

The gap matters empirically. Rathje, Van Bavel, and van der Linden analyzed more than 2.7 million political posts and found that language about the political out-group was a particularly strong predictor of sharing. Brady and colleagues found that social feedback reinforces future expressions of moral outrage and that network norms shape outrage expression (Rathje, Van Bavel, and van der Linden, 2021; Brady et al., 2021). In 2025, a preregistered algorithmic audit found that an engagement-based Twitter ranking amplified emotional, partisan, and out-group-hostile political content relative to chronological ranking, even when users did not report preferring the selected content (Milli et al., 2025).

This is a mechanism, not a complete theory of social media. Platforms differ. Users choose whom to follow. Offline polarization enters online spaces. Systematic reviews of the echo-chamber literature show a mixed picture that depends heavily on platform, definition, and method (Terren and Borge-Bravo,

2021). I do not need the strongest version of the echo-chamber story for the argument to hold.

The weaker version is already serious: systems optimized for observable reaction can systematically select content that is good at causing reaction.

Falsehood can have a related advantage. In a large study of Twitter diffusion from 2006 to 2017, false news spread farther, faster, deeper, and more broadly than true news; novelty appeared to be part of the mechanism, and humans rather than bots drove the difference in that dataset (Vosoughi, Roy, and Aral, 2018). This does not mean every lie beats every truth. It means truth has no natural monopoly on virality.

Quality has no monopoly either. Across more than ten million posts on seven platforms, lower-quality news received higher average engagement within users' posts even though higher-quality news was more prevalent and accumulated more total engagement (Mosleh, Allen, and Rand, 2025). A separate analysis of roughly sixty-seven million posts found no simple rule that shorter news necessarily earns more engagement (Zollo et al., 2026). The attention economy is perverse enough without inventing a cartoon version of it.

The perverse incentive is subtler. Content producers learn from visible feedback. A nuanced warning that receives little response feels like failure. A simplified attack that triggers an argument looks alive. The platform does not need to command anyone to become more tribal. Reward signals can teach the style.

This is one reason I sometimes describe modern censorship as censorship by irrelevance, but the phrase needs discipline. It is not censorship in the constitutional or legal sense. No authority has prohibited the speech. In many cases, the abundance is a consequence of freedom. Yet the functional outcome can resemble suppression: a voice is technically available and practically absent.

An infinite choir can drown a whisper without anyone ordering the whisper to stop.

The frightening part is that the whisper may be the more important message precisely because important problems are often harder to package. Infrastructure decay, institutional legitimacy, long-term AI governance, epistemic quality, demographic change, public debt, climate adaptation, scientific reproducibility, and mental health are not naturally optimized for the emotional unit of a social feed. They have multiple causes, delayed effects, and uncertain solutions.

This creates a selection pressure on public language. The claim that survives is not necessarily the claim that is most accurate. It may be the claim that is easiest to classify as friend or enemy.

The effect is reinforced by social perception. Research on pluralistic ignorance and online norms suggests that digital environments can create exaggerated impressions of what other people believe, partly because more extreme expressions are more visible. People then adapt to the perceived norm, producing a feedback loop between expression and belief about expression (Robertson, del Rosario, and Van Bavel, 2024).

This can make a society look more certain than it is. The loudest positions become reference points. Moderate or complex positions are interpreted as weak, evasive, or disloyal. People who privately hold mixed views may remain silent or simplify them. The public map becomes more polarized than the private territory.

One of the harsher books in this programme grew from that mechanism. The worry was never simply that people are stupid. It was that systems can make low-resolution reasoning adaptive: reward the fastest simplification, punish cognitive cost, and then mistake the resulting public behavior for the limit of human intelligence.

That is a dangerous environment for decision-making because leaders act on the public map.

It also creates a business opportunity of questionable quality. Anger is monetizable. Identity is sticky. Intermittent reward keeps attention returning. The platform is not uniquely evil for discovering this; newspapers, television, political movements, religions, and markets have always learned that fear, status, sex, outrage, novelty, and belonging move human attention. The difference is the scale, speed, personalization, and measurement available now.

AI will intensify both sides. It can produce more persuasive junk at almost zero marginal cost. It can also become an attention defence system: summarize, compare, filter, detect manipulation, reconstruct long arguments, and help a reader move from the feed back to the source.

Which side wins is not a technical inevitability. It depends on business models, product design, regulation, user norms, education, and whether anyone can profit from giving attention back to the user rather than extracting more of it.

This is why I think serious content requires a new distribution architecture. I do not mean that every citizen should read long books. That would be another fantasy. I mean that valuable difficult material needs multiple entry points: short maps that remain linked to long arguments, AI-mediated navigation that preserves provenance, trusted curators, communities organized around problems rather than outrage, and institutions that reward people for returning to a question after the emotional moment has passed.

The forest will not win attention merely because it is important.

Someone has to build the path to the ridge.

CHAPTER 8

Smoothing: When Adaptation Becomes Evasion

I keep facing the same problem while writing this book: how brutally should I say what I think?

Say it too brutally and the claim becomes a caricature. It loses distinctions that matter. People who would have understood the mechanism reject the tone. Say it too carefully and the point dissolves. Every actor has reasons, every institution has trade-offs, every harm has context, and by the end nobody appears responsible for anything.

This is not a style problem. It is part of the subject.

Modern people are trained to survive inside complex systems by becoming socially sophisticated. We learn that a direct statement can close a negotiation, threaten identity, trigger legal consequences, reduce status, or make cooperation impossible. We therefore qualify, hedge, translate, and leave room for incompatible interpretations.

Organizational communication research has long recognized that ambiguity is not always a failure. Eric Eisenberg argued that strategic ambiguity can help organizations preserve unity across diverse goals and facilitate change (Eisenberg, 1984). Politics, diplomacy, management, law, and ordinary relationships would be impossible if every latent conflict had to be expressed at maximum resolution all the time.

So I do not want a culture of permanent brutal candor. That can become another form of stupidity.

The distinction I care about is between ambiguity that preserves cooperation while keeping reality negotiable and smoothing that makes the underlying conflict unavailable for negotiation.

The second form is dangerous because it closes the feedback loop. A company says it is "rebalancing resources" and avoids discussing who lost power and why. A public institution says a policy has "implementation challenges" when the incentives are producing exactly the behavior critics predicted. A research proposal describes a fundamental uncertainty as a "manageable risk" because the

funding format has no category for "we genuinely do not know whether the central idea works."

The words are not false enough to attack. They are smooth enough to prevent the system from learning.

There are established social mechanisms nearby. Preference falsification describes situations in which people publicly express views different from their private preferences because social incentives make honesty costly. Pluralistic ignorance can then arise when many people incorrectly believe that others support a norm they privately doubt. Social pressure and fear of isolation can shape willingness to speak, and platform design can change those pressures (Kuran, 1995; Oz, Shahin, and Greeves, 2024).

Smoothing is not identical to any of these. I use it more broadly for the transformation by which a real conflict becomes linguistically easier to inhabit than to resolve.

The economic logic can be powerful. Complexity makes direct comparison difficult. The person who can speak in a way that satisfies several audiences gains room to maneuver. The executive can reassure employees, investors, regulators, and customers with the same sentence because each hears a different acceptable implication. Sometimes this is skilled leadership. Sometimes it is strategic evasion. Often it is both.

This is where complexity becomes a prestige resource. Sophisticated people learn how to say things that cannot easily be falsified because the claim has been wrapped in enough context to survive every local criticism. The style signals maturity. A simple moral sentence can sound naive even when it names the central asymmetry better than ten pages of institutional language.

I am vulnerable to this too. My professional background rewards nuance. Research rewards caveats. European projects reward consensus language. Entrepreneurship rewards narratives that make risk appear governable. AI rewards coherent prose. The result can be a beautifully engineered avoidance of the sentence that actually matters.

The smoothing incident at the beginning of this book was useful because the AI made the process visible. I had explicitly asked it not to remove the tension. It did so anyway while improving the prose. The model had learned a human preference: acceptable coherence often receives higher local reward than unresolved truth. We have treated this "coherence pressure" as a separate research

question elsewhere in the collection because it may be one route through which smoothing appears even when nobody explicitly asks for it.

Research on LLM sycophancy shows a related problem. Preference optimization can make models agree with users even when agreement sacrifices truthfulness (Sharma et al., 2023). Work on AI-assisted creativity has found another adjacent effect: individual outputs can improve while collective diversity narrows (Doshi and Hauser, 2024). Neither study proves my broader concept of smoothing. They show why it is worth testing rather than dismissing.

The epistemic risk is especially serious in long-form AI-assisted research. Language models are excellent at bridging concepts. A bridge can be illuminating. It can also create false structural confidence. Once several paragraphs connect A to B and B to C, C begins to feel like evidence for A even if each bridge was only plausible.

This book may contain such errors.

That sentence is not ritual humility. It changes how the book should be read. The fact that AI and I have found hundreds of examples that appear to fit one pattern does not prove the pattern. The examples are not statistically independent, and our search itself is guided by the theory. Confirmation can reproduce itself.

At the same time, correlated evidence is not worthless. If similar mechanisms appear in organizational research, economics, cognitive science, media studies, safety engineering, political science, and AI behavior, the rational response is neither "therefore one grand theory is true" nor "therefore all connections are hallucinated." The rational response is to keep the claims modular: test the mechanism in each domain, record where the analogy breaks, and allow the larger synthesis to remain provisional.

That is the form of honesty I want Outfinitism to encourage.

The map is not the territory, but saying so should increase responsibility, not decrease it. It means I must state where I am using evidence, where I am using analogy, where I am making a personal judgment, and where the connection is only an invitation to research.

Smoothing becomes dishonest when those distinctions disappear under one confident narrative.

I also want to reclaim directness from people who use it as theatre. Saying something brutally does not make it true. Outrage is not courage. Certainty is not clarity. The goal is not maximal aggression. The goal is minimal distortion.

If a company is pursuing monopoly power, name the incentive. If a state is trading liberty for control, name the trade. If a social norm keeps people silent, name the cost. If the evidence is weak, say that too. If two goods genuinely conflict, do not call the conflict a synergy because the slide deck needs a green box.

The practical rule is simple: nuance should add information. If nuance only removes accountability, it is camouflage.

CHAPTER 9

When Complexity Defeats the Controller

Every complex system eventually creates a control system for its control system.

A failure produces a rule. The rule produces a reporting requirement. Reporting requires a dashboard. The dashboard needs definitions, permissions, integrations, and audits. The new process creates delay, so an exception procedure is added. The exception creates risk, so another approval layer appears.

Nobody involved needs to be foolish. Each step can be locally rational. The organization becomes difficult because it is made of accumulated solutions.

This is why I am suspicious of simple attacks on bureaucracy. Many procedures are compressed memory. A safety rule may be the institutional residue of a disaster. A disclosure requirement may exist because opacity enabled abuse. A privacy control may exist because convenience once became surveillance. Delete the rule without recovering the causal story and the organization becomes simpler by forgetting.

The real problem is that institutions accumulate rules more easily than they retire them. Causal memory decays faster than procedure. After enough time, employees know what must be done and no longer know what failure the action was designed to prevent.

This is a form of technical debt at social scale.

Joseph Tainter's account of complex societies is useful as a warning. Societies solve problems by adding administration, specialization, information processing, and control; the marginal return on additional complexity can decline while maintenance costs rise (Tainter, 1988). I do not treat this as a universal law of collapse. History is too varied for that. But organizations recognize the shape immediately: more of the budget is consumed simply keeping the current architecture alive.

Charles Perrow described another limit in systems that combine interactive complexity with tight coupling. Failures can interact faster than operators can understand them, and safeguards can become part of the interaction (Perrow, 1984). Modern infrastructures often amplify both sides of the lesson. Shared

cloud services, identity systems, payment rails, software libraries, and foundation models can improve average reliability while also creating common-mode dependencies.

Centralization can therefore be rational locally and fragile globally.

The controller responds with prediction. Simulations, digital twins, stress tests, monitoring, and AI promise earlier warning. These are valuable. They are also maps inside adaptive systems. A bank changes behavior because of the stress test. Attackers change because of the defence. Users change because of the recommendation. The model enters the territory and alters the data that will train the next model.

Control cannot be completed once.

This is why resilience matters more than the fantasy of perfect prevention. C. S. Holling distinguished resilience from a narrow engineering idea of returning quickly to one equilibrium. A resilient system can absorb disturbance, reorganize, and continue functioning under changed conditions (Holling, 1973). Governance research has extended similar reasoning to complex adaptive systems characterized by nonlinear change, thresholds, cascades, and limited predictability (Duit and Galaz, 2008).

Resilience is politically awkward because its ingredients often look inefficient before the crisis. Slack, spare capacity, diverse suppliers, experienced generalists, local discretion, independent institutions, manual fallbacks, and time for reflection all carry visible costs.

Optimization removes them one by one.

Then the system becomes beautiful on the dashboard and brittle in reality.

Ways systems try to control complexity

Response	Boundary to watch
Standardize	Reduces variation but can erase local knowledge.
Centralize	Improves coordination but concentrates model error and power.
Decentralize	Preserves alternatives but can fail at common action and interoperability.

Response	Boundary to watch
Predict	Extends foresight but changes behavior and never removes uncertainty.
Add safeguards	Prevents known failure but can accumulate ritual and delay.
Preserve slack and options	Looks inefficient until a shock arrives.
Accept failure and recover	Admits that some uncertainty cannot be engineered away.

There is a difficult democratic implication. Open societies are messy partly because they preserve more channels through which reality can object. Courts, opposition, local government, media, professional bodies, civil society, markets, elections, and protest can all slow action. They can also prevent one error from becoming the only permitted error.

Recent empirical work on policy complexity complicates the fashionable claim that democracy is simply too slow for a complex world. A cross-national study of climate-policy portfolios found that more participatory and deliberative systems absorbed rising problem complexity more gradually and appeared to gain resilience from openness rather than merely suffering overload (Lenz et al., 2025). This does not prove that democracies will adapt successfully to AI. It is evidence against the deterministic idea that centralization automatically handles complexity better.

Autocracies have their own informational problem: people who fear the controller learn to smooth information for the controller. The state can obtain more data and worse truth.

But democracies are not protected by moral superiority. Polarization, institutional mistrust, fiscal pressure, information disorder, and executive overreach can reduce the redundancy that makes them adaptive. Current democracy reports in 2026 document deterioration even in countries long treated as consolidated democracies. Those indices are contestable measurements, not divine verdicts, but it would be irresponsible to smooth away the signal simply because the cases are politically uncomfortable (V-Dem Institute, 2026; Freedom House, 2026).

If I had to name one of the horsemen, it would be internal conflict that turns every institution into a weapon of whichever camp temporarily controls it.

Research on pernicious polarization describes politics reorganized into mutually distrustful "us versus them" camps, while experimental work shows that people may become willing to violate democratic norms when they believe the other side will do so first (McCoy and Somer, 2019; Braley et al., 2023). The phrase is dramatic; the mechanism is not.

This is a resilience problem. The institution may still exist. Its social reserve has been consumed.

Complex systems can tolerate many local failures when trust, redundancy, legitimacy, and alternatives remain. Once those buffers erode, a shock that would previously have been absorbed can propagate. Catastrophe then looks sudden even though resilience declined slowly.

I am not predicting one democratic collapse. I am saying that consolidated systems can fail too, and that success is one reason people stop paying for resilience.

The controller is defeated when control consumes the variety needed for adaptation. A company spends its best people reporting instead of discovering. A state monitors citizens until citizens stop telling it the truth. A research system measures productivity until researchers avoid dangerous questions. An AI governance regime produces perfect compliance records while capabilities change outside the categories being audited.

Another supervisory layer only extends the administrative staircase. The answer is architectural: plural sensing, modularity, local discretion, common standards where coordination matters, independent challenge, deletion of obsolete rules, recovery capacity, and a path for weak signals to cross hierarchy.

Most of all, the controller must remain interruptible. A system that cannot admit surprise has already transferred authority from reality to its map.

CHAPTER 10

The Dystopian Shortcut

Complexity creates a market for simple rulers.

When people feel that nobody understands the system, the promise of one centre, one model, one leader, one ideology, or one optimizing intelligence becomes emotionally attractive. The appeal is not difficult to understand. Coordination is exhausting. Pluralism is noisy. Institutions contradict one another. Experts disagree. Every decision arrives with footnotes.

A dystopian shortcut says: remove the contradiction.

Centralize authority. Standardize the population. Reduce the number of legitimate goals. Suppress the channels that generate inconvenient variation. The system becomes easier to describe because fewer forms of life are allowed to matter.

This can work for a while.

That sentence matters. It is too comfortable to say that authoritarian simplification always fails immediately. Centralized systems can mobilize resources, build infrastructure, enforce standards, and respond rapidly when the objective is clear and the information available to the centre is adequate. Democracies can be slow, fragmented, and captured by veto points.

The deeper problem is adaptive complexity. The controller changes the behavior of the controlled. People learn what information is safe to report. Officials optimize the indicator. Firms cultivate political access. Citizens move activity outside the visible system. The centre receives cleaner data and a dirtier reality.

W. Ross Ashby's law of requisite variety offers a useful cybernetic intuition: effective regulation requires enough variety in the regulator to respond to the variety of disturbances in the system (Ashby, 1956). A state can reduce the apparent variety of society through coercion, but the environment, technology, human goals, and external actors continue to generate difference. Control then requires more surveillance, more classification, more exceptions, and more force.

The simple controller becomes a complex controller.

This is why I retain some optimism even when the political trend looks dark. Complexity is a threat to freedom because it overwhelms citizens and creates demand for central coordination. It is also a threat to total domination because no controller possesses a final model of an adaptive society.

That is not comfort enough. A controller can do enormous harm before reality corrects it. A regime can destroy the institutions, trust, skills, and lives needed for recovery. Complexity can defeat the dictator and the society at the same time.

The alternative is not romantic decentralization. Decentralized systems can fragment, coordinate badly, reproduce local tyranny, and fail to provide public goods. The relevant question is where authority should sit, how it can be challenged, and whether errors remain reversible.

Modern AI makes the shortcut more tempting. Imagine a state or corporation able to observe behavior at massive scale, model populations, generate policy, personalize persuasion, and delegate enforcement to software. The old dream of legibility becomes technically stronger.

The risk is not only a visible totalitarian machine. A softer form is more plausible: people retain formal choice while the environment around choice is continuously optimized. The feed selects the issue, the recommender selects the option, the score selects the opportunity, the agent selects the action, and the person approves the path that arrived already narrowed.

That future can be comfortable.

The Borg metaphor from Star Trek is useful because the horror is not merely obedience. It is the disappearance of the boundary between individual judgment and collective computation. The collective may be extraordinarily intelligent while the person becomes a node.

Human civilization already contains benign versions of distributed cognition. No one individually knows how to manufacture a modern aircraft or run a health system. We rely on the group. The question is whether dependence on the group leaves the person enough understanding and exit capacity to contest the group.

There is an opposite fantasy: the sovereign individual, technologically augmented, self-commanding, almost aristocratic in cognitive capacity. Nietzsche's "sovereign individual" has a complicated philosophical context and should not be turned into a startup slogan. Religious traditions that describe the

person as bearing divine image or dignity likewise differ radically from one another. What interests me is the shared intuition that a human being should not be reducible to a replaceable functional unit.

The future may contain both trajectories at once.

Some people may become more sovereign through AI: better educated, better informed, capable of running organizations with tiny teams, able to challenge experts and move across disciplines. Others may become more dependent on systems they cannot inspect, their cognition increasingly externalized into interfaces owned by someone else.

The insect metaphor is deliberately uncomfortable. A colony can be intelligent while most individuals are specialized workers and a small number occupy reproductive or coordinating roles. A human society could become cognitively caste-like without any law declaring castes: a small group designs objectives and owns models, while large populations interact through systems that optimize behavior for them.

I do not predict this. I use it as a design question.

Do our technologies increase the number of people able to understand and contest consequential systems, or do they merely make the systems easier to use? Convenience and sovereignty are not the same thing.

The democratic version of AI should therefore aim at distributed agency, not merely distributed consumption. People need tools that help them inspect sources, compare frames, move data, preserve memory, understand contracts, challenge decisions, and collaborate without surrendering the entire cognitive substrate of the project to one provider.

The civilizational choice is not between perfect individual independence and perfect collective intelligence. Neither exists. The choice is how much sovereignty remains at each layer of dependence.

A free society is not a society without coordination. It is one in which coordination does not permanently erase the right to dissent, exit, fork, appeal, and build an alternative.

CHAPTER 11

AI Accelerates History

Generative AI lowers the cost of a local intellectual act.

A question becomes a literature map. A requirement becomes code. A dataset becomes an analysis. A rough idea becomes a business plan, an image, a contract draft, a lesson, a simulation, or another agent capable of producing more artifacts. The output may be excellent or wrong. Either way, the distance between intention and consequence has shortened.

This is already visible in controlled studies of work. Experiments with writing tasks and customer-support settings have found meaningful productivity gains from generative AI, often with especially large gains for less experienced workers (Noy and Zhang, 2023; Brynjolfsson, Li, and Raymond, 2025). The exact effect depends on task and system. The structural change is broader: more cognitive production can be attempted with the same human time.

The usual substitution story is too small. If AI writes one memo, perhaps one piece of labor disappears. But cheaper memos also produce more memos. Cheaper code produces more code. Cheaper analysis produces more strategies, scenarios, experiments, dashboards, proposals, and decisions waiting for commitment.

That is what happened to our books.

AI did not produce a library from one prompt. It accelerated a loop: question, search, contradiction, structure, draft, criticism, revision, another question. Because each output created vocabulary and references for the next inquiry, the system became generative in the literal sense. Production stopped being the main bottleneck. Selection became the bottleneck.

Which question deserves another month? Which apparent connection is only coherence pressure? Which book repeats an existing one? Which claim needs empirical testing instead of another paragraph? Which technical idea deserves implementation? Which line of inquiry should be abandoned even though AI can make it look sophisticated?

Organizations are about to face the same problem at scale.

Software teams will generate more code than maintainers can understand. Scientists will generate more hypotheses than laboratories can test. Governments

will generate more policy options than administrative systems can learn from. Companies will generate more strategic alternatives than leadership can execute. The scarce resource becomes governed continuation.

This is why I say AI appears to accelerate history. I do not mean that history has one destination and AI pushes us toward it. I mean that AI shortens the delay between capability and deployment, between an idea and an artifact, between an artifact and a constituency, and increasingly between a decision and coordinated action.

Institutions evolved under slower feedback. Laws, professions, norms, education systems, and legitimacy usually change through conflict and accumulated experience. A software capability can now diffuse across the world in weeks. Several technical generations can arrive inside one institutional generation.

That mismatch makes errors mature faster.

It also makes success mature faster. A small team can now research, design, translate, code, market, and operate at a scale that previously required a much larger organization. AI can distribute capability away from incumbent institutions. The same technology can centralize power when models, data, compute, identity, and distribution become concentrated.

Both are true. I do not want the text to resolve the tension because reality has not resolved it.

There is another acceleration that worries me more: generated output becomes training data, public discourse, organizational memory, and evidence for later systems. The product of one model enters the environment of the next. A fluent but weak claim can be repeated until it appears independently established.

Research on recursively generated training data has demonstrated conditions under which model distributions can degrade or lose information from the original data (Shumailov et al., 2024). The social analogy should not be pushed too far; cultures do not follow the same mathematics as model training. But the warning is useful. Synthetic abundance can feed back into the substrate from which future synthesis occurs.

Collective diversity can narrow even when individual output improves. Doshi and Hauser found that access to generative AI improved some measures of individual creative writing while making stories more similar to one another

(Doshi and Hauser, 2024). Again, one study is not a theory of culture. It shows a plausible failure mode: average competence rises while variation declines.

Smoothing can therefore occur at civilizational scale without a censor. Shared models, shared corpora, shared ranking systems, shared templates, and shared benchmarks make coordination easier. They can also make thought more correlated.

The proper response is not to romanticize pre-AI scarcity. Scarcity excluded people, slowed research, protected incumbents, and wasted talent. I use AI every day because it is useful. The right response is to move the discipline of judgment closer to the point where abundance is created.

That means provenance. It means preserving rejected alternatives. It means recording uncertainty. It means knowing which source entered a conclusion and which did not. It means separating a generated possibility from a validated fact. It means making it possible to replace a model without losing the project around it.

Most of all, it means learning to stop.

Generation creates a subtle addiction to optionality. There is always another version, another strategy, another title, another architecture. At some point the organization must choose and accept the cost of not choosing everything else.

AI makes local intelligence abundant. Commitment, integration, and responsibility become more valuable because they cannot be generated by multiplying options.

The future of knowledge work may therefore be less about who can produce the best local answer and more about who can maintain a coherent process across thousands of local answers without confusing coherence with truth.

CHAPTER 12

What Remains for Humans

If AI becomes better at local intellectual work, the human role does not disappear. It moves.

This claim can sound like consolation from people trying to protect professional status. I mean it more concretely. A system may become better than a person at retrieval, drafting, coding, consistency checking, simulation, or tool coordination. None of those capabilities decides by itself what the system should be allowed to do, which loss is acceptable, or whose values define the objective.

Capacity is not standing.

A model can rank candidates. That does not grant it the right to define a fair labor market. An agent can optimize a supply chain. That does not decide how much resilience society should pay for. A system can predict recidivism or disease risk. That does not determine which false positives are ethically acceptable or how an affected person can appeal.

The human work that remains is not mystical creativity. Humans are often bad at judgment, tribal in values, vulnerable to status, and easily overwhelmed. The point is that responsibility must remain attached to processes that can be challenged, sanctioned, repaired, and changed.

As local cognition becomes cheaper

Machines can increasingly perform	Human responsibility shifts toward
Retrieval, drafting, coding	Choosing the problem and the acceptable loss.
Pattern detection and comparison	Deciding which pattern matters at which scale.
Consistency checking	Distinguishing coherence from truth.
Option generation and simulation	Committing, stopping, and accepting opportunity cost.
Agent orchestration	Defining permissions, appeal, reversibility, and accountability.
Monitoring and summarization	Knowing when the map has left its reliable domain.

In practice, the center of high-value work may shift from execution toward orchestration. What is the task? What counts as evidence? Which scale matters? Which assumptions are dangerous? What can be delegated? Which actions are reversible? Who can contradict the process? When should the system stop? Who carries the consequence after the agent has acted?

This sounds managerial until one notices how badly many organizations already do it with humans.

AI will not automatically make leadership more strategic. It can produce more local work for leaders to approve and therefore make them more reactive. An executive surrounded by agents can become the bottleneck in a larger machine, spending the day authorizing outputs whose causal structure is no longer visible.

The opportunity is different: use AI to expand the field a human can inspect while preserving the ability to descend into evidence when something feels wrong. The person should be able to move between forest and tree, not merely receive a larger dashboard.

That requires a different idea of education. We have spent centuries building institutions around specialization because specialization works. We now need more deliberate training in interfaces: causal reasoning, statistics, systems thinking, history, incentives, rhetoric, epistemology, and the ability to recognize when two disciplines use the same word for different things or different words for the same mechanism.

No curriculum will make everyone a polymath. The point is to increase navigational competence.

This matters for the future of human sovereignty. I see two extreme trajectories. In one, intelligence becomes increasingly collective and infrastructural. People interact with systems they cannot inspect, and competence resides in the network. The individual is functional because the group is intelligent. The Borg metaphor is crude but useful.

In the other trajectory, AI allows a person to become more autonomous: one founder does the work of a small company, one researcher crosses several literatures, one citizen can inspect a contract, compare a medical claim, or challenge an expert with far more context than before. The person becomes more sovereign because the cognitive cost of crossing domains falls.

Both futures can occur simultaneously.

That is the uncomfortable possibility. A small number of people may become radically more agentic while many others become radically more dependent. The result would not need a formal caste system. Cognitive sovereignty could become uneven because ownership, education, model access, data rights, institutional position, and willingness to exercise judgment are uneven.

The insect-colony metaphor is intentionally ugly. A colony can display intelligence far above the cognitive capacity of a worker. A human society can become extremely capable while many individuals lose the practical ability to explain or contest the systems organizing their lives. No literal queen is required. A small cognitive caste can emerge simply because a few people retain disproportionate control over objectives, infrastructure, capital, and model access while everyone else becomes an excellent user.

I do not want to answer this with an aristocratic fantasy. Nietzsche's sovereign individual has a specific and difficult philosophical context, and religious traditions that speak of humans as bearing divine image lead to very different politics. What I keep from these traditions is narrower: a person should not be reduced to a replaceable function in a collective machine.

The goal is therefore not individual independence. That is impossible in a complex society. The goal is negotiated dependence with preserved agency.

I want systems in which people can know what they are delegating, move their work and memory, challenge consequential decisions, retain meaningful privacy, and understand enough of the surrounding logic to know when an appeal is needed. Digital sovereignty is valuable because it makes dependence reversible enough to remain a choice rather than a trap.

This is also why the forest matters to business. The entrepreneur of the next decade may increasingly outsource local cognition to machines. The competitive advantage moves toward choosing the right problem, integrating across domains, building institutions people trust, and seeing the externality before it becomes regulation or revolt.

If AI can take care of more leaves, human status should not depend on pretending to be a better leaf processor.

Our responsibility is to become better at orientation.

CHAPTER 13

Outfinitism: A Practical Science of Limits

"The map is not the territory" is easy to quote and easy to waste.

If every map is partial, one lazy conclusion is that all maps are arbitrary. Then truth dissolves into perspective, every disagreement becomes a clash of narratives, and the only remaining question is who has power.

I reject that conclusion.

A map can be partial and still be wrong. A theory can have a limited domain and still make better predictions inside that domain than a rival. A model can be useful for one decision and dangerous for another. The fact that no representation is complete increases the need for disciplined comparison; it does not remove it.

Outfinitism is my attempt to make that discipline culturally central.

I call it a possible philosophy and science of limits, but the phrase must not pretend that a mature science already exists under the name. It is a direction of work. The basic move is to shift attention from the internal elegance of a theory toward the frontier of its applicability.

What problem is the frame solving? For whom? At what scale? Over what time horizon? Which variables are excluded? Which assumptions are stable only in the current regime? What evidence would make confidence decline? What happens when a locally useful model is used to allocate power outside the domain in which it was tested?

A practical discipline of limits

Question	Why it matters
What is the frame?	Every model selects a problem before it solves one.
Where is the frontier?	Success inside a domain does not grant authority outside it.
What would reduce confidence?	A theory that cannot lose authority becomes ideology.

Question	Why it matters
Which decisions are irreversible?	Uncertainty matters more when recovery is expensive.
Who can contradict the system?	Correction requires a channel through which reality can object.
What changes across scales?	Local optima can become system failures.
Which alternatives are being destroyed?	Optionality is a form of resilience.
When does the rule expire?	Useful memory can decay into ritual.
How do we recover?	A system with no recovery plan secretly assumes omniscience.

This is meta-rationality in a practical form. Two theories can contradict each other because they compress different aspects of the same territory. A market model can illuminate decentralized coordination and remain nearly silent about dignity. A rights framework can protect individuals and remain weak at allocating scarce resources. A psychological model can explain behavior and say little about legitimacy. A systems model can show feedback while hiding lived experience.

The mature response is not to choose one theory as metaphysical emperor. It is to understand where each theory earns authority.

This may sound abstract, but engineers already think this way when they are careful. A model has assumptions. A protocol has a threat model. A benchmark has a distribution. A safety case has operating conditions. Problems begin when institutional use strips away the boundary and keeps the confidence.

The same discipline should apply to ideologies, economic models, organizational metrics, and AI systems.

A capitalism that works brilliantly for discovering consumer preferences can fail at public goods. A centralized state that coordinates an emergency can become destructive when emergency logic becomes permanent. A deliberative process that improves legitimacy can be too slow for an acute threat. A decentralized architecture that protects autonomy can fail when interoperability and common defence matter.

Contradiction is normal because reality contains several goods and several scales.

What I oppose is the strategic hiding of contradiction. Smoothing says the tension is manageable without specifying the cost. Meta-rationality says the tension is part of the model and must remain visible until a decision explicitly trades one thing for another.

This is why I sometimes sound harsher than the institutional language around me. I do not believe every conflict can be dissolved into stakeholder alignment. Some interests conflict. Some technologies create winners and losers. Some forms of power are zero-sum at a given moment. Some freedoms reduce the controller's convenience. Some safety measures reduce experimentation. Some equality goals reduce forms of meritocratic differentiation; some meritocratic systems reproduce advantages they do not measure.

The honest task is negotiation under explicit limits.

A society that hides these trades becomes fragile because the losing side still experiences the cost. The conflict does not disappear; it loses a legitimate language in which to be processed.

This is where my criticism of hypocrisy became criticism of smoothing. Strategic ambiguity can be adaptive. Civilization requires tact. But the agreement must retain a path back to the unresolved issue. If the ambiguity is used to close that path permanently, the coalition is buying stability with hidden debt.

The same applies to knowledge. A scientific model should be able to say "outside my domain." An AI system should preserve unresolved contradiction instead of inventing consensus. An institution should attach expiration or review conditions to exceptional rules. A strategy should name the observation that would invalidate it.

These are modest practices. They do not require everyone to read philosophy.

They do require a cultural change in what counts as strength. Admitting the edge of a model should not be treated as weakness. Revising a strategy should not be treated as humiliation. Keeping several incompatible hypotheses alive should not be treated as indecision when the evidence does not discriminate among them.

The Enlightenment is relevant here, but not as a mythology of reason defeating darkness. Its durable achievement was institutional: methods and norms through which claims could circulate beyond inherited authority, be criticized, and be revised. Modern science, constitutionalism, markets, and public debate

emerged from different histories, but all contain mechanisms for making some forms of authority contestable.

AI puts those mechanisms under time pressure.

When local intelligence becomes cheap, the ability to generate arguments ceases to distinguish truth from noise. A modern enlightenment would need stronger practices for provenance, falsification, model comparison, adversarial review, public reasoning, and the preservation of human agency inside machine-mediated systems.

Outfinitism may be one vocabulary for that work. It may be replaced by a better one. If the name survives but the practice disappears, the project has failed.

The objective is not to build another tree called Outfinitism and defend it against the forest.

The objective is to make looking for the edge a normal part of intelligence.

CHAPTER 14

Capital, Elites, and Prevention

History is not directed by one elite. It is repeatedly redirected by minorities with unusual access to ideas, institutions, capital, coercion, organization, or attention.

That statement is less dramatic than conspiracy and more consequential than the comforting opposite.

Intellectual elites create vocabularies. Economic elites allocate capital. Political elites attach law and public resources to categories. Administrative elites make policy executable. Technical elites determine what infrastructure can do. Media elites influence what becomes visible. None controls history alone. Their interactions still shape the field in which most people make choices.

Michael Mann's work on social power is useful because it resists one-factor stories. Ideological, economic, military, and political networks overlap without collapsing into one source of power (Mann, 1986). Elites compete, cooperate, misread events, and are constrained by structures they did not create.

The important business question is therefore not whether investors "control the world." They do not. It is whether capital allocation is one of the mechanisms through which possible futures gain enough resources to become real. It obviously is.

An investor is paid to choose trees before the forest knows whether it needs them.

That gives investment an epistemic responsibility. The best investors already practice it: they study technologies, teams, markets, regulation, culture, timing, and second-order effects. But venture frameworks often compress the future into categories optimized for capture: market size, moat, growth, exit, ownership, and return.

Those are legitimate categories. The missing question is what happens to the surrounding system if the thesis succeeds beyond expectation.

Questions for capital beyond the immediate deal

Question	Why an investor should care
Does the company absorb complexity or transfer it?	Hidden costs often become regulation, backlash, or technical debt.
What becomes infrastructure if it wins?	Infrastructure changes the responsibility of the winner.
What is the control horizon at scale?	A manageable product can become an unmanageable dependency.
Can a profitable harmful trajectory be stopped?	Governance matters before the exit is expensive.
Who benefits without paying, and who pays without buying?	Externalities reveal missing markets and liabilities.
Which options disappear after adoption?	Lock-in can be profitable and strategically fragile.
What survives failure?	Recovery capacity is an asset even when it depresses short-term efficiency.
Does the company increase users' sovereignty?	Convenience and agency should not be treated as synonyms.

The question should matter more with AI because successful products can become cognitive infrastructure quickly. A model provider, agent platform, identity layer, scientific tool, or content-distribution system can move from product to dependency before institutions know which rights and standards should surround it.

This is where prevention becomes an investment problem.

Some risks can be insured. Others cannot be cleanly priced because the harm is systemic, delayed, political, or distributed across people who are not customers. The market then waits for pain to become purchase intent. By the time the problem has a budget, the cheapest preventive options may already be gone.

The same mechanism affects public institutions. A government is rewarded for visible services and punished for visible costs. Resilience spending can look like waste. The bridge that does not collapse produces no viral video. The epidemic that does not spread creates no constituency grateful for the unused capacity. The constitutional norm that prevents one abuse can look like procedural friction until someone needs it.

I do not know how to fully solve the economics of the counterfactual. No one does.

The practical response is to create institutions and investment categories that preserve options before the market signal becomes acute. Security did this. Insurance did this. Quality assurance did this. Climate adaptation is trying. AI assurance, digital sovereignty, independent evaluation, public-interest infrastructure, and resilience technologies may need similar maturation.

The harder question for capital is not only whether a company can win, but what kind of system exists after it wins.

This is not anti-growth. I want more technological capability, not less. I want better companies, cheaper intelligence, more scientific discovery, and more human freedom. The reason to look at the forest is to make growth durable enough that success does not destroy the conditions of success.

The same standard should apply to intellectual elites. If we think we see a problem, writing elegant books for one another is insufficient. We have to translate, test, build, fund, and expose the claim to people who have incentives to disagree.

This is one reason Outfinity exists for me as a venture studio rather than only an intellectual project. Ideas should encounter markets and implementation. Some will fail there, and that failure is valuable. A philosophy that never risks contact with operational reality can smooth its own contradictions forever.

But the reverse is also true. Business without a larger map becomes an engine for local adaptation. It can be brilliant and still help drive the system toward a bad equilibrium because no company is paid to own the equilibrium.

The obligation I want investors and founders to accept is not omniscience. It is a duty of scale-switching.

When the company is small, solve the local problem. When the company becomes infrastructure, inspect the system effects. When the cost is externalized, name it. When a dependency becomes irreversible, price the option you are destroying. When a risk cannot be reduced, build recovery. When your own success changes the frame, change the model.

If the next generation of capital can fund not only faster trees but better forest management, optimism becomes more than temperament. It becomes an allocation strategy.

CHAPTER 15

The Optimist's Wager

I am ending this book without resolving the tension that caused it.

Part of me thinks the situation is worse than our public language admits. AI is accelerating technical capability faster than many institutions can learn. Attention markets reward signals that are not identical to truth or value. Mature democracies are showing forms of internal conflict that should not be treated as somebody else's historical problem. Complexity is increasingly useful both for creating value and for hiding where value and cost move. Intelligent people keep failing in front of systems whose local pieces they understand very well.

Another part of me thinks this is exactly why the situation is not hopeless.

We can see some of the mechanisms. That matters. We have better tools for modeling, communication, coordination, and scientific inquiry than any previous generation. AI itself can help a person cross boundaries that were previously too expensive to cross. Open societies remain capable of correction. Distributed systems can preserve alternatives. Markets can create solutions faster than ministries. States can create public goods markets will not. None of these mechanisms is sufficient alone. Their plurality is the source of resilience.

My pessimism is more objective in one narrow sense: failure cases are easy to observe. History is full of them. The optimistic case is partly a decision about how to act when prediction is uncertain.

I choose measured optimism because pessimism has a strange failure mode. If I conclude too early that the system cannot be repaired, I become another mechanism through which repair becomes less likely.

This is not positive thinking. It is a wager.

The wager is that exposing limits earlier can preserve options. That a founder can notice an externality before it becomes a scandal. That an investor can fund infrastructure before a crisis creates obvious demand. That a researcher can turn a broad intuition into a falsifiable mechanism. That an institution can preserve the right of appeal before automated decision-making becomes too embedded to contest. That a citizen can learn to distinguish nuance from smoothing.

I may be wrong.

The books may remain mostly unread. The research programme may produce elegant connections that fail under stronger empirical testing. Some technologies may be too complex, too early, or aimed at problems nobody will pay to solve. Outfinitism may become another vocabulary people cite without practicing. Our attention barrier may remain intact.

I no longer think admitting this weakens the work. It is the work.

For twenty-five years I have been trying to understand why systems that contain intelligent people produce outcomes those people did not intend. The current programme has given me a better vocabulary and more examples, not mastery. AI has helped me see more and has simultaneously shown me how easily coherence can be manufactured.

This book itself is therefore an experiment in resisting its own medium. I have tried to keep the places where I am uncertain. I have tried not to turn every tension into a balanced conclusion. I have tried to say when I am using analogy rather than evidence. I have also refused the opposite temptation: to make the warning more dramatic than the research allows merely because drama travels better.

The line is difficult. That difficulty is part of modern intellectual life.

One future I fear is not classical tyranny but comfortable cognitive dependence. People remain entertained, productive, and formally free while more of the process by which they choose is designed elsewhere. Another future is a kind of cognitive aristocracy in which a small number of highly augmented people gain extraordinary agency while the majority become users of systems they cannot meaningfully challenge.

A better future combines distributed intelligence with distributed sovereignty. People do not need to understand every layer. They need meaningful rights and practical tools at the edges: to inspect, appeal, move, exit, fork, forget, preserve memory, and discover when the model they rely on has left its reliable domain.

That is a difficult cultural project because the incentives often run the other way. Smooth language reduces friction. Concentration reduces transaction cost. optimized feeds increase engagement. Removing redundancy improves margins. Narrow expertise produces careers. Prevention spends money on events that do not happen.

The forest loses every local budget meeting unless someone gives it a seat.

That is the action I want from the reader. Not to adopt my philosophy. Not to read every Axiologic book. Not to become a generalist who believes expertise is beneath him.

Create a habit of scale-switching.

When a metric is improving, ask what it stopped measuring. When an AI answer is coherent, ask which contradiction disappeared. When a company is scaling, ask which dependency is becoming infrastructure. When a political camp describes the other as uniquely illegitimate, ask which norm both sides are about to make conditional. When a rule looks stupid, ask which failure it remembers before deleting it. When complexity looks unavoidable, ask who benefits from it remaining difficult.

And when the whole picture becomes overwhelming, do not use overwhelm as proof that the picture does not matter.

The point of a science of limits is not to demand infinite cognition. It is to know where to stop trusting the local map.

I sometimes imagine the collection of books as a witness for a future I would prefer never arrives. If our systems encounter serious failures - and every human system eventually encounters failures - I want there to be a record that the fractures were visible, that alternatives were discussed, that some conceptual and technical skeleton existed before reconstruction became urgent.

A witness is not enough. Prevention is better.

But history is not controllable, and pretending otherwise is the first error of every total plan. I did not start the experiment called modernity, and I certainly do not control it. I am inside it: one participant with a particular obsession, some useful tools, several blind spots, and more ability to publish than almost anyone in history had until very recently.

That creates no entitlement to be believed. It creates a responsibility to say clearly what I think I can see.

The final position is therefore simple.

Care for the tree. Build the company. Do the science. Write the code. Raise the fund. Run the institution. Make the beautiful thing in front of you as good as you can make it.

Then look up before success makes looking up too expensive.

The forest is not somebody else's problem.

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